Structural Bayesian Techniques for Experimental and Behavioral Economics

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November, 2025

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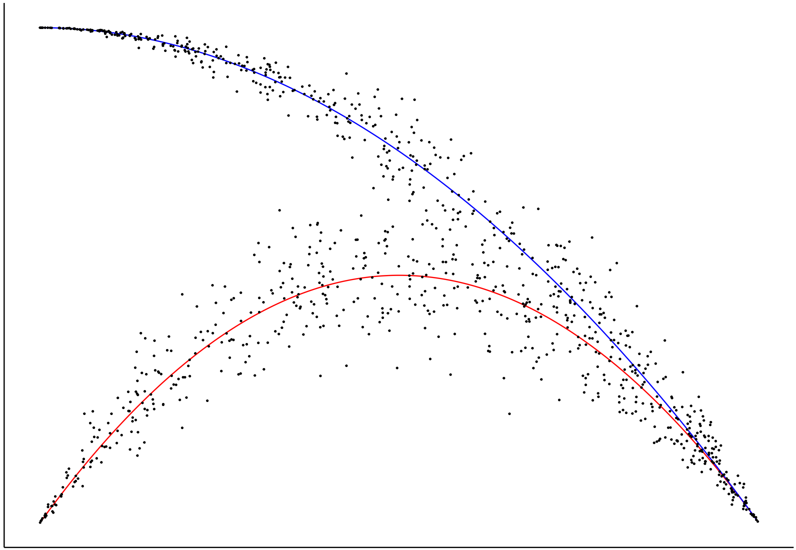


Introduction

Answering research questions with structural econometrics

I have a model and data, and I want to ...

- **Estimate the parameters** in the model
- Compute **economically meaningfully quantities** from the parameters in my model
- **Quantify how much heterogeneity** is present between participants in my model's parameters
- Use my model to make a **prediction** about behavior in another setting

But you should also **quantify the uncertainty** you have in all of these

Estimating parameters in a model

My model says that parameter(s) are important, so I want to know what they are.

For example, some preference models

$$\text{Risk: } U(L) = E\left(\frac{X^{1-r}}{1-r} \mid L\right)$$

$$\text{Time: } U(x) = \sum_{t=0}^{\infty} \delta^t x_t$$

$$\begin{aligned}\text{Other-regarding: } U_i(x) = x_i &- \frac{\alpha}{n-1} \sum_{j \neq i} \max\{x_j - x_i, 0\} \\ &- \frac{\beta}{n-1} \sum_{j \neq i} \max\{x_i - x_j, 0\}\end{aligned}$$

Computing economically meaningful quantities

What I really want to do is interpret a **transformation** of the parameters.

E.g. a certainty equivalent:

$$U(L) = E\left(\frac{X^{1-r}}{1-r} \mid L\right)$$

$$C(L) \sim L$$
$$C(L) = \left[E\left(X^{1-r} \mid L\right)\right]^{\frac{1}{1-r}}$$

$C(L)$ converts expected utility units (hard to interpret and compare) into dollars

Quantifying heterogeneity

Parameter θ is likely *not* the same for every participant

Assume (for example):

$$\theta_i \sim iidN(\mu, \sigma^2)$$

Estimate and interpret μ and σ

Making predictions

Parameters θ imply something about choices y in scenario S:

$$p(y \mid \theta, S)$$

- For the same participant in another scenario
  - Model evaluation
- For different participants in similar scenarios
  - Scaling an experiment up from a pilot
- For different participants in a new scenario
  - Designing a new experiment

What could these models look like?

A **utility function** our participants could be maximizing

- E.g. risk preferences

An **equilibrium concept** in a game

- E.g. Quantal response equilibrium

A set of economically-relevant **decision rules**

- E.g. Strategy frequency estimation

–>

Structural estimation

Take an **economic model** of how data are generated, and **estimate the parameters** in this model

Then:

- Interpret the estimated parameters
- Derive implications of the estimated parameter values

Comparison with traditional reduced-form experimental procedures

Theory says that choices $y \mid D$ will (or will not) be different depending on treatment conditions D.

Design an experiment to test whether or not $y \mid D_1$ has the same mean, distribution, etc. as $y \mid D_2$

Compare treatment outcomes, e.g. with linear regression:

$$y_{i,t} = \beta_0 + \beta_1 D_{i,t} + \epsilon_{i,t}, \quad H_0 : \ \beta_1 = 0$$

Here's the differece

Reduced-form:

- Model says something about $y \mid D$
- Estimate/test for differences between treatments

Structural:

- Model **is** a distribution $y \mid \theta, D$
- Estimate θ

θ tells us how treatment conditions D affect choices y

We will draw a straight line through the data only if this is what the model says the data will look like!

The structural approach is *very* different to traditional experimental techniques

Not necessarily a need for:

- Randomization into treatments
- Any (between-subjects) treatments at all
- Designs that could falsify a theory

But you can still use data from a traditionally-designed experiment.

- Structural estimation is a *complement* to the traditional experimental approach
- But: traditional experiment designs may not be optimized for structural estimation

Structural models for economic experiments

I have a model that describes how participants will make decisions in my experiment.

I will teach you about **probabilistic** models, which specify a probability distribution over decisions in the experiment.

- **Likelihood** $p(y \mid \theta)$: If you tell me the model's parameters, I can simulate data for my experiment.
- **Prior** $p(\theta)$: I have beliefs about what these parameters will look like before running the experiment

Together, I can simulate data for my experiment that is consistent with my prior beliefs about my model's parameters

$$p(y) = \int_\Theta p(y \mid \theta)p(\theta)\mathrm{d}\theta = E_\theta\left[p(y \mid \theta)\right]$$

Applying Bayes' theorem to learn from data

Bayes' theorem in general:

$$p(A, B) = p(A \mid B)p(B) = p(B \mid A)p(A)$$

therefore: $$p(A \mid B) = \frac{p(B \mid A)p(A)}{p(B)}$$

Bayesian statistics: $A = \theta$ (parameters), $B = y$ (data):

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)} \propto p(y \mid \theta)p(\theta) = p(y, \theta)$$

$p(y \mid \theta)$ is the **likelihood**: given θ, what will my data look like?

$p(\theta)$ is the **prior**: what are my beliefs about θ before running my experiment?

Why Bayes?

This is **the right way** to learn from data, according to math!

Why Bayes? More practically

The alternative is (usually) maximum likelihood.

- Bayes often turns a difficult maximization problem into a slightly less difficult simulation problem.
- The technology to do this simulation has improved *immensely*

Better handling of un-observable heterogeneity

- Data augmentation vs. integrating the likelihood
- Handled particularly well by Hamiltonian Monte Carlo (This is what *Stan* does)
- Participants are heterogeneous!

Applying Bayes' rule to structural models in experiments

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)} \propto p(y \mid \theta)p(\theta)$$

A typical procedure:

$u(y; \theta, x)$: A (deterministic) **utility function** over choices y given experimental conditions x and parameters θ

$p(y \mid u, \theta)$: A **stochastic choice rule**

$p(\theta)$: a prior for our model's parameters

Some components of a structural model: Utility function

If present, this is usually the thing we are most interested in

$u(y;\theta,x)$: A (deterministic) **utility function** over choices y given experimental conditions x. For example:

Risk preferences:

$$u(y;\theta,x)=\sum_{k=1}^{K}p_k^y\frac{x_k^{1-r}}{1-r}$$

Time preferences:

$$u(y;\theta,x)=\sum_{t=0}^{\infty}\delta^t y_t$$

The components of a structural model: A probabilistic choice rule

Converts a **deterministic** prediction into a **probabilistic** one.

- Specifies a probability distribution over actions given utility of taking these actions
- Necessary for constructing a likelihood
- Behaviorally plausible

$$\text{logit choice: } p(a \mid x, \theta) = \frac{\exp(\lambda u_a(x, \theta))}{\sum_{j \in A} \exp(\lambda u_j(x, \theta))}$$

$$\text{optimal plus error: } y \mid x, \theta \sim \arg\max_{y' \in A} u(y', x, \theta) + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

The components of a structural model: A prior

$$p(\theta)$$

Our belief about θ before we observe our data

What are:

- Likely values?
- Unlikely but possible values?
- Stupidly unlikely but still possible values?
- Impossible values?
- Values which pose a problem for estimation?

A **formal mathematical statement** of this

How can we estimate these models?

The posterior distribution $p(\theta \mid y)$ is in general not known in closed form.

- Fortunately, we can *sample* from it.
- We then report *summary statistics* of these simulated draws (e.g. means, standard deviations).

The more "traditional" Bayesian approach, Gibbs sampling: needs exact conditional distributions for subsets of the parameters.

- Great if you want to only work with linear models (I don't)
  - Conjugate priors
- Feasible for (mostly non-linear) structural applications, but very slow (see my job market paper (Bland 2019))
- More **user-intensive**

A better way for estimating structural models

Most of the computationally difficult stuff is in handling the non-linearities, which do not work well with a Gibbs sampler.

Instead, use a *probabilistic programming language* that takes care of all of this!

I will teach you about *Stan* (Carpenter et al. 2017):

- Free, and callable within *R*, *Python*,
- Allows us to code up models in a similar way to how we might write them down
- Uses Hamiltonian Monte Carlo: Fast!
- Especially well suited to handling hierarchical models
- In-built diagnostics

Example *Stan* program for estimating a Normal distribution

```
data {
  int<lower=0> N; // number of observations in data
  vector[N] y; // data
  vector[2] prior_mu; // prior for parameter mu (mean and s
  real<lower=0> prior_sigma; // prior for sigma (half-cauch
}
parameters {
  real mu; // mean of distribution
  real<lower=0> sigma; // standard deviation of distributio
}
model {
  y ~ normal(mu, sigma); // likelihood contribution
  mu ~ normal(prior_mu[1],prior_mu[2]); // prior for mu
  sigma ~ cauchy(0.0,prior_sigma); // prior for sigma
}
```

What information does *Stan* need? **Overview**

Instructions on how to compute the log joint density of parameters and data:

$$\log p(y, \theta) = \underbrace{\log p(y \mid \theta)}_{\text{likelihood}} + \underbrace{\log p(\theta)}_{\text{prior}}$$

This is called the `target`

What information does *Stan* need? **Data block**

A description of the data (and any other information you want to pass to *Stan*)

- names
- data types, e.g. `int`, `real`, `vector`, etc.
- array sizes
- bounds (optional)

```
data {
  int<lower=0> N; // number of observations in data
  vector[N] y; // data
  vector[2] prior_mu; // prior for parameter mu (mean and s
  real<lower=0> prior_sigma; // prior for sigma (half-cauch
}
```

What information does *Stan* need? **Parameters block**

- names
- bounds
- parameter types
  - Must be continuous variables
  - `real`, `vector`, `matrix`, `simplex`, `corr_matrix` etc. allowed
  - `int` not allowed

```
parameters {
  real mu; // mean of distribution
  real<lower=0> sigma; // standard deviation of distributio
}
```

What information does *Stan* need? **Model block**

Instructions on how to compute the `target`:

$$\log p(y, \theta) = \log p(y \mid \theta) + \log p(\theta)$$

```
model {
  y ~ normal(mu, sigma); // likelihood contribution
  mu ~ normal(prior_mu[1],prior_mu[2]); // prior for mu
  sigma ~ cauchy(0.0,prior_sigma); // prior for sigma
}
```

Alternatively:

```
model {
  target += normal_lpdf(y | mu, sigma); // likelihood contr
  target += normal_lpdf(mu | prior_mu[1],prior_mu[2]); // p
  target += cauchy_lpdf(sigma | 0.0,prior_sigma); // prior
}
```

Optional *Stan* blocks

`functions`: Specify user-defined functions

`transformed data`: Data manipulations

`transformed parameters`: Calculate functions of parameters and data that can be used in the `model` block (slow-ish)

`generated quantities`: Calculate functions of parameters and data that cannot be used in the `model` block (fast)

Running *Stan* in *R*

See `exampleStan.R` and `exampleStan.stan`

Some other things to know about *Stan*

Diagnostics:

- Only about four things that can go wrong with Hamiltonian Monte Carlo
- All of them diagnosable (*Stan* does this for you)
- Extensive documentation on them and how to fix them

Speed:

- Compiles from `C++` $\implies$ very fast once compiled
- Speed improved through vectorization
- Parallelization supported (both within- and between-chain)

Will I need a super-fancy computer to do this?

No, but you *will* need a bit of patience

Everything I show you today could be done done on my laptop (bought in 2017) overnight

- Microsoft Surface Pro running Windows 10 Pro
- Processor Intel(R) Core(TM) i7-7660U CPU @ 2.50GHz, 2496 Mhz, 2 Core(s), 4 Logical Processor(s)
- 16 GB RAM (didn't need anything near this)
- *R* version 4.2.1 (2022-06-23 ucrt)
- *RStan* version 2.26.13 (*Stan* version 2.26.1)

Noticeable slow-down of other processes while running this. That's why I also have a my office computer

Outline for the rest of this workshop

1. Models with optimization
   - Estimating a model of other-regarding preferences
2. Hierarchical models
   - Unobservable, participant-specific heterogeneity
3. Doing something with the posterior estimates
   - Generated quantities and out-of-sample predictions
4. Conclusion

Models with optimization

What are our participants maximizing?

Economic models usually involve some kind of **optimization**

One way to represent this is with a utility function:

$$\max_{y \in \mathbb{Y}} U(y; \theta, x)$$

We can lean about θ, the parameters in $U(\cdot; \cdot, \cdot)$, by observing choices y in different experiment conditions x.

We will often have to make an assumption about the functional form of $U(\cdot; \cdot, \cdot)$.

With an estimate of the parameters in U we can

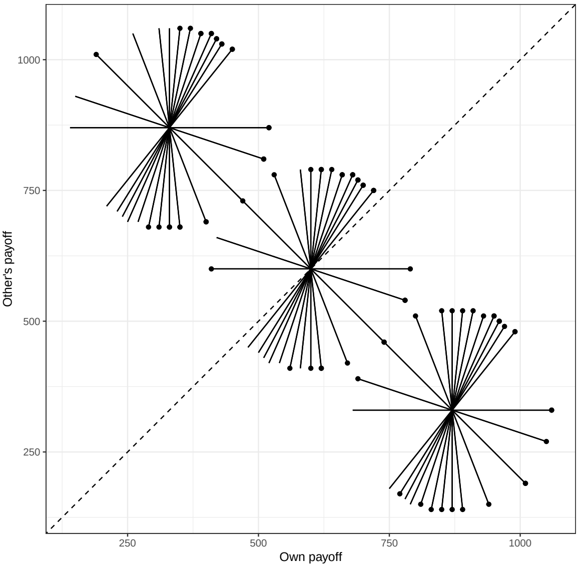
$$\max_{y \in \mathbb{Y}} U(y; \theta, x)$$

- **Interpret** the estimated parameters θ
- Calculate **economically meaningful quantities** that are functions of θ
- Make (deterministic) **predictions** of what participants might do in different experiment conditions x and with different choice sets $\mathbb{Y}$

Example dataset: Bruhin, Fehr, and Schunk (2019)

- 174 participants
- 78 pair-wise choices each
- Allocations of money between self and other participant

Bruhin, Fehr, and Schunk (2019)



A model: Fehr and Schmidt (1999)

$$U_i(x, y) = x - \alpha_i \max\{0, y - x\} - \beta_i \max\{0, x - y\}$$

where: $x = i$'s earnings

$y =$ the other person's earnings

Parameters:

- α_i: Aversion to disadvantageous inequality
  - How much I dislike the other person getting more than me
- β_i: Aversion to advantageous inequality
  - How much I dislike the other person getting less than me

Fehr and Schmidt (1999) argue that $\alpha_i \geq \beta_i$ and $0 \leq \beta_i \leq 1$

What can we do with this model?

$$U_i(x, y) = x - \alpha_i \max\{0, y - x\} - \beta_i \max\{0, x - y\}$$

Given parameter values for α_i and β_i, we can:

- Predict which of the two allocations will be chosen

We cannot:

- Simulate data (because the model is deterministic)
- Therefore we can't estimate our model (yet)

The problem(s) with deterministic choice

Economic models often assume *perfect* optimization ("rationality"), but ...

Knowing our participants:

- Real people make mistakes
- Real people don't have exactly the utility functions we want to assume they have

From a technical perspective:

- Mistakes + deterministic model = zero likelihood!

For a variety of reasons, we need a *probabilistic* model.

Probabilistic models

Model implies a *probability distribution* over actions in the choice set.

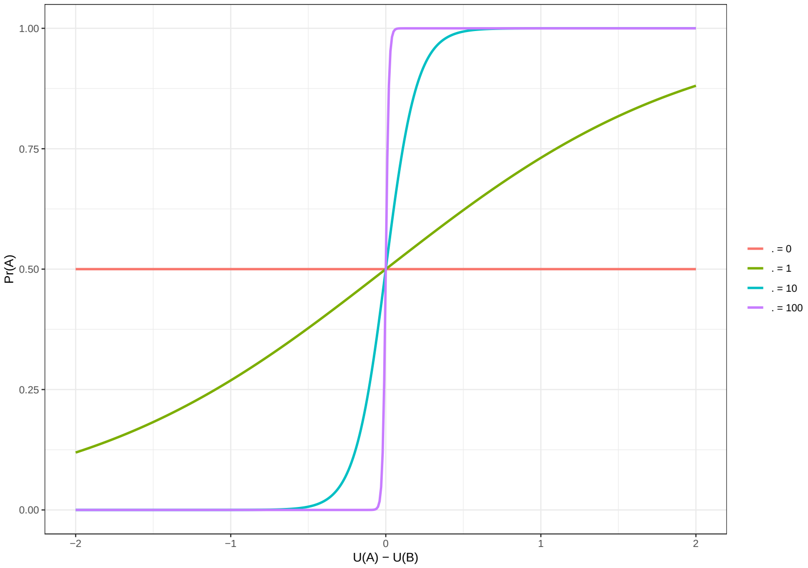
Nice property to have: the probabilistic model looks like noisy optimization

A good place to start: logit choice

$$\Pr(A \mid U_i(A), U_i(B), \lambda) = \frac{\exp(\lambda U_i(A))}{\exp(\lambda U_i(A)) + \exp(\lambda U_i(B))}$$

- $\lambda \geq 0$: choice precision
- $50/50$ randomization when $\lambda = 0$
- Approaches noiseless optimization as $\lambda \to \infty$
- Options with greater utility are more likely to be chosen than options with less utility

Logit choice



Up to this point

A (deterministic) model of utility

$$U_i(x, y) = x - \alpha_i \max\{0, y - x\} - \beta_i \max\{0, x - y\}$$

A probability distribution over actions:

$$\Pr(A \mid U_i(A), U_i(B), \lambda) = \frac{\exp(\lambda U_i(A))}{\exp(\lambda U_i(A)) + \exp(\lambda U_i(B))}$$

At this point we can:

- Simulate data given parameters $(\alpha_i, \beta_i, \lambda_i)$
  - This is *really useful* when designing an experiment!
- Estimate the model using MLE (if we really want to)

To do Bayesian inference, we need (good) priors!

What is a prior?

Qualitatively: what we know about our parameter(s) before we see the data

As experts (yes, you are one), you should be able to make a reasonable judgement about what are likely and unlikely realizations of a parameter

- A daily discount factor of $\delta = 0.95$ is ridiculously small (but not for yearly)
- Relative risk-aversion of 0.5 (i.e. square-root utility if CRRA) is not too surprising
- $\beta = 1,000$ in Fehr and Schmidt (1999) is a stupidly large amount of aversion to advantageous inequality.

But we have to **formally state** this in a **mathematical** expression of the uncertainty we have in our parameter(s)

Step 1: picking a distributional family

Choose a family of distributions that has the right support.

E.g. $\lambda \geq 0$, choice precision:

- Normal – nope: can't have negative numbers
- Binomial – nope: doesn't cover $\mathbb{R}_+$
- Log-normal – good

$\alpha, \beta \in \mathbb{R}$:

- normal

Step 2: Calibrate the parameters of the distribution

$$
\begin{aligned}
\alpha_i &\sim N(m_\alpha, s_\alpha^2) \\
\beta_i &\sim N(m_\beta, s_\beta^2) \\
\log \lambda_i &\sim N(m_\lambda, s_\lambda^2)
\end{aligned}
$$

Translate our (maybe not very well-formulated) beliefs into $\{m_\alpha, s_\alpha, \ldots\}$

This is usually the hard part, think about:

- What parameter values seem Likely? Plausible? Unlikely? Just plain stupid?
- What parameter values make plausible predictions?
- What parameter values might pose problems for estimation?

For α and β

Let's place 75% of the prior probability mass of both parameters in the range $(-1, 1)$, which means that our prior distributions for α and β are:

$$\alpha \sim N(0, 0.67^2)$$
$$\beta \sim N(0, 0.67^2)$$

I.e.

$$\Pr\left[N(0, 0.67^2) \in (-1, 1)\right] = 0.75$$

For λ

What do different values of λ say about decisions?

To make the math easy, look at the predictions for a selfish ($\alpha = \beta = 0$) participant:

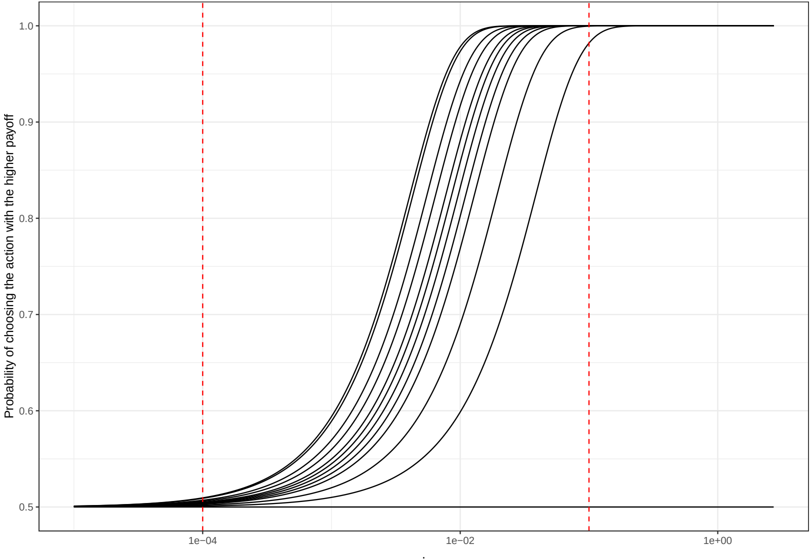
The unique selfish payoff differences in the experiment are:

```
d<-(D
    |> mutate(absDiff = abs(self_x-self_y))
    |> dplyr::select(absDiff)
    |> unique()
    )

d |> as.vector() |> print()
```

```
## $absDiff
##  [1] 140 200 380 240 120  80  40 280 180   0 160 360
```

Selfish behavior and λ



λ calibration

Set prior 5th percentile to 0.0001, 95th percentile to 0.1

$$
\begin{aligned}
\log(0.1) &= m_\lambda + 1.64 s_\lambda \\
\log(0.0001) &= m_\lambda - 1.64 s_\lambda \\
\log(0.1) - \log(0.0001) &= 2 \times 1.64 s_\lambda \\
s_\lambda &= \frac{\log(0.1) - \log(0.0001)}{2 \times 1.64} \approx 2.11 \\
m_\lambda &= \log 0.1 - 1.64 \times 2.11 \approx -5.76
\end{aligned}
$$

Now we can estimate the model!

(So let's look at the code, `RepAgentModels_BFS2019.stan`)

Model estimates – pooled

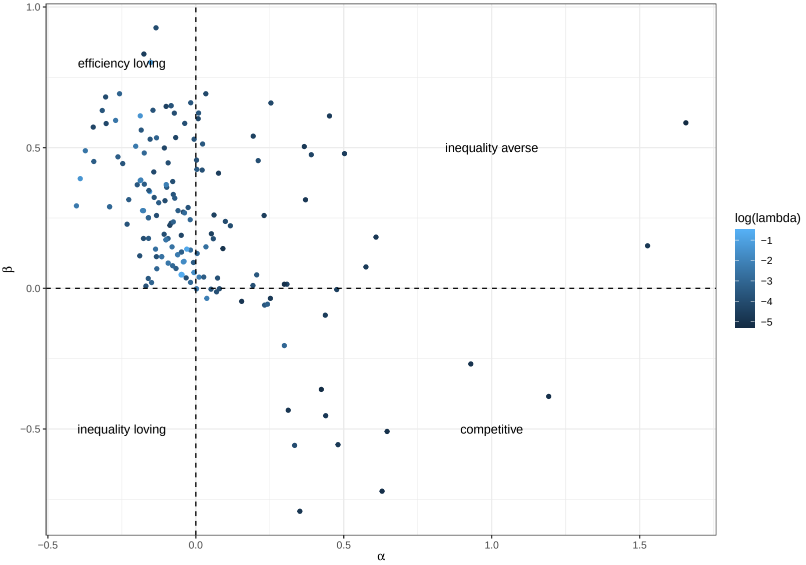
Stan summary output

				perce		
	mean	se_mean	sd	X2.5.	X25.	
alpha	-0.051	0.00	0.008	-0.065	-0.056	-(
beta	0.262	0.00	0.008	0.247	0.257	(
lambda	0.016	0.00	0.000	0.015	0.015	(
lp__	-4022.763	0.03	1.236	-4025.964	-4023.325	-402

Not particularly interesting – participants aren't homogeneous

Model estimates – participant-specific

(showing posterior means only)



Hierarchical models

Taking heterogeneity seriously

- Our participants are likely heterogeneous in many dimensions
  - Observable (i.e. post-experiment survey answers)
  - Unobservable (i.e. different model parameters)

Here we are going to model the **unobservable** heterogeneity

We might be:

- Interested in this heterogeneity
- Worried about heterogeneity (e.g. heterogeneity bias, e.g. Wilcox (2006))
- Seeking better estimates of individual participants' parameters that leverage the entire dataset

Treat participants' parameters as random draws from a population

$$\theta_i \doteq \begin{pmatrix} \alpha_i \\ \beta_i \\ \log \lambda_i \end{pmatrix} \sim iidN(\mu, \Sigma)$$

Now need to specify "hyper-priors" for:

- μ: mean of participants' (transformed) parameters
- Σ: variance-covariance matrix

μ and Σ become informative priors for θ_i.

Joint estimation of $\{\mu, \Sigma\}$ and $\{\theta_i\}_{i=1}^N$

Joint estimation?

$$\theta_i \doteq \begin{pmatrix} \alpha_i \\ \beta_i \\ \log \lambda_i \end{pmatrix} \sim iidN(\mu, \Sigma)$$

Estimate individual-level parameters $\{\theta_i\}_{i=1}^{N}$ alongside population-level parameter μ and Σ

$$\begin{aligned} p(\theta, \mu, \Sigma \mid y) &\propto p(y \mid \theta, \mu, \Sigma) p(\theta, \mu, \Sigma) \\ &= \underbrace{p(y \mid \theta)}_{\text{likelihood}} \overbrace{p(\theta \mid \mu, \Sigma)}^{\text{prior}} \underbrace{p(\mu, \Sigma)}_{\text{hyper-prior}} \end{aligned}$$

Likelihood: Same as individual-level estimation!

Hyper-priors

Decompose Σ into a vector of standard deviations τ and a correlation matrix Ω

$$\begin{aligned} \mu_\alpha &\sim N(0, 0.39^2), & \tau_\alpha &\sim \mathrm{Cauchy}^+(0, 0.79) \\ \mu_\beta &\sim N(0, 0.39^2), & \tau_\beta &\sim \mathrm{Cauchy}^+(0, 0.79) \\ \mu_\lambda &\sim N(-5.76, 1^2), & \tau_\lambda &\sim \mathrm{Cauchy}^+(0, 0.24) \\ \Omega &\sim \mathrm{LKJ}(2) \end{aligned}$$

Similar reasoning to individual participant model.

Estimating the model in *Stan*

(see `Hierarchical_BFS2019_correlated.stan`)

Things to note:

- Likelihood calculation is exactly the same as in individual-level estimation
- Joint estimation of individual-level parameters

Here I follow the recommendations of https://mc-stan.org/docs/stan-users-guide/regression.html#multivariate-hierarchical-priors.section

- Decomposition of Σ into standard deviations τ and correlation matrix Ω
- Cholesky factorization of Ω
- Non-central parameterization of individual-level parameters

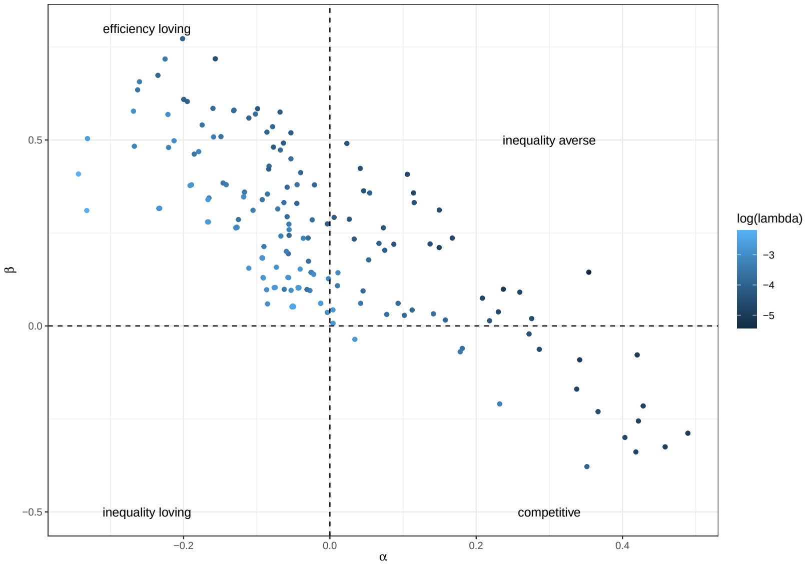
Things we can do with a hierarchical model

1. Comment on the estimates of **central tendency** μ ("representative agent")
2. Comment on estimates of **spread** Σ
3. Estimate individual-level parameters with a prior **informed by the entire dataset**
   - Anything we can do with individual-level estimates we can also do with a hierarchical model!

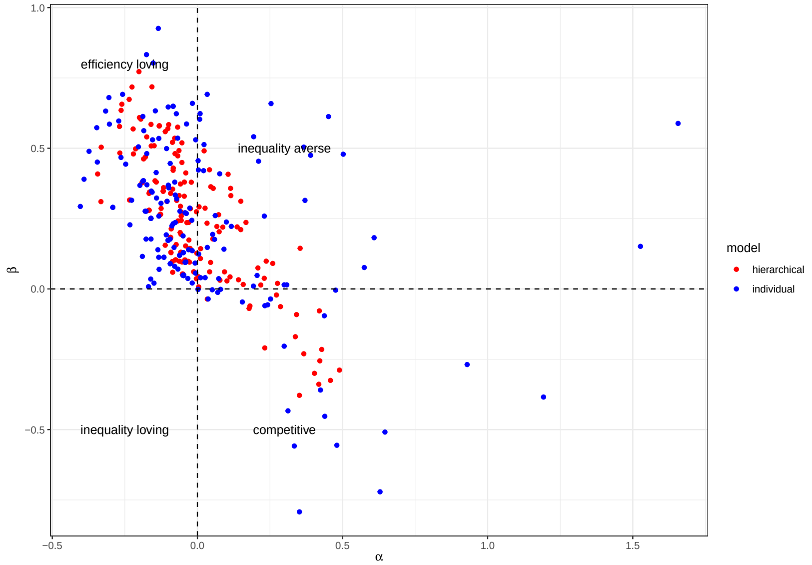
Population-level estimates

parameter	mean	se_mean	sd	X2.5.	X25.	X50
μ_α	-0.014	0.001	0.015	-0.044	-0.024	-0.014
μ_β	0.230	0.001	0.020	0.191	0.216	0.230
μ_λ	-3.531	0.002	0.068	-3.662	-3.576	-3.532
τ_α	0.175	0.000	0.014	0.149	0.165	0.174
τ_β	0.242	0.000	0.016	0.211	0.231	0.241
τ_λ	0.774	0.002	0.062	0.660	0.731	0.771

Individual-level estimates



Comparing the hierarchical and individual models



Up to this point

We can estimate the parameters of our model.

However: Sometimes the raw parameters are less interesting than *functions* of the parameters:

We might be interested in:

- Predictions
  - Out-of-sample
  - Designing a new treatment or experiment
- Welfare measures
- Classifying participants into groups

Beyond estimation: `generated quantities`

Sometimes we are interested in *transformations* of our parameters

- We estimate a CRRA utility function, but we are really interested in a risk premium
- We estimate a quantal response equilibrium, but we are really interested in its prediction for another game
- We estimate the fraction of participants who bracket decisions narrowly, but we are really interested in how much utility they are giving up (my JMP).

We are often **more** interested in these transforms than the parameters themselves

The good news (and there's no bad news)

We get transformations of parameters for free!

Once we have our posterior simulation, to get a simulation of transformations of parameters, we just transform the simulated parameters.

E.g. If $\{\theta_s\}_{s=1}^{S}$ are our simulated parameters, and $g(\theta)$ is the transformation we are interested in, then:

$$E[g(\theta) \mid y] \approx \frac{1}{S} \sum_{s=1}^{S} g(\theta_s)$$

(the approximation is up to simulation accuracy)

Even better: we can do this in *Stan*, so it all happens very fast!

Application: predicting behavior in another environment

A new task:

You have 1000 tokens. You will earn 1 experimental currency unit per token that you choose to keep. You can also buy income for another participant (who has no tokens) at a price of ρ tokens per experimental currency unit. That is, if you spend 100 tokens on the other participant, they earn $\frac{100}{\rho}$ experimental currency units.

Research question: What is the population demand for giving to the other participant?

We already have a model for this

What is the population demand for giving to the other participant?

$$U_i(x, y) = x - \alpha \max\{0, y - x\} - \beta_i \max\{0, x - y\}$$

Budget constraint: $1000 = x + \rho y$

and logit choice over the budget set (discretized to units of 1 token)

Solving the model ...

Substituting in the budget constraint gets us:

$$U_i(x) = x - \alpha_i \max\left\{0, \frac{1000 - x(1+\rho)}{\rho}\right\} - \beta_i \max\left\{0, \frac{x(1+\rho) - 1000}{\rho}\right.$$

Then logit choice gives us:

$$\Pr(X = x \mid \alpha_i, \beta_i, \lambda_i, \rho) = \frac{\exp(\lambda_i U_i(x))}{\sum_{k=0}^{1000} \exp(\lambda_i U_i(k))}$$

and the expected amount kept is:

$$E[X \mid \alpha_i, \beta_i, \lambda_i, \rho] = \sum_{x=0}^{1000} x \frac{\exp(\lambda_i U_i(x))}{\sum_{k=0}^{1000} \exp(\lambda_i U_i(k))}$$

. . . more working . . .

But what we really want is Y, so using the budget constraint:

$$E[Y \mid \alpha_i, \beta_i, \lambda_i, \rho] = \frac{1000 - E[X \mid \alpha_i, \beta_i, \lambda_i, \rho]}{\rho}$$

This is the expected quantity demanded given parameters $\{\alpha_i, \beta_i, \lambda_i\}$.

From here, we want to take a population-average over these parameters, so applying the law of iterated expectations:

$$E[Y \mid \rho] = E\left[E[Y \mid \alpha_i, \beta_i, \lambda_i, \rho] \mid \rho\right]$$

This is a function of the population-level parameters μ and Σ. It is hard to evaluate analytically, but we can approximate it numerically . . .

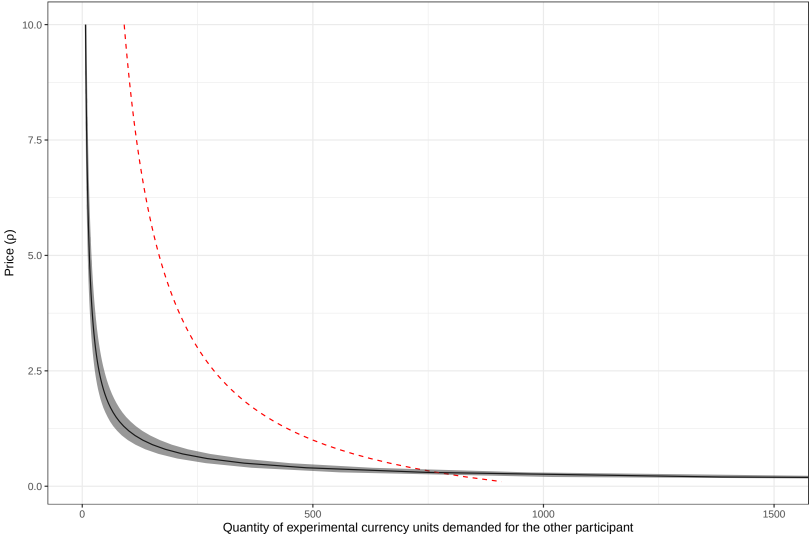
. . . more working

$$E[Y \mid \rho] = E\left[E[Y \mid \alpha_i, \beta_i, \lambda_i, \rho] \mid \rho\right] \approx \frac{1}{S}\sum_{s=1}^{S} E[Y \mid \tilde{\alpha}_s, \tilde{\beta}_s, \tilde{\lambda}_s, \rho]$$

$$\text{where: } \begin{pmatrix} \tilde{\alpha}_s \\ \tilde{\beta}_s \\ \log \tilde{\lambda}_s \end{pmatrix} \sim iidN(\mu, \Sigma)$$

(go back to *Stan* model `Hierarchical_BFS2019_correlated.stan`)

Results



Red line shows the equal split
Shaded region shows a 95% Bayesian credible region, 2.5th−97.5th percentiles.

Conclusion

Big picture

I have a model and I want to:

- Estimate and comment on the parameters
- Estimate and comment on *transformations* of the parameters
- Assign a measure of uncertainty to all of the above

We have a framework for doing this

Bayesian statistics

Applying Bayes' rule to quantify posterior uncertainty in:

- Our model's parameters

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

- Assigning probabilities to models

$$p(M \mid y) \propto p(y \mid M)p(M)$$

The structural approach

Estimate the parameters in the model

- Interpret the parameters
- Interpret implications of the parameters, e.g.:
  - Predicted actions
  - Certainty equivalents

Compared to reduced-form:

- Compare differences in treatments,
- Draw a straight line through the data, interpret the slope

Heterogeneity

Our participants are heterogeneous, and we should model this!

Bayesian techniques do this well

Some useful tools:

- Hierarchical models: Continuous types
  - Data augmentation
- Mixture models: Discrete types
  - Integrate the likelihood

Heterogeneity in a parameter could be important, even if you don't care about that parameter!

So when should I think about structural?

Before running an experiment!

- What parameters/quantities do I want to estimate?
- What might my data look like?
- Can I modify my experiment to better answer my research question?
- Will my experiment adequately estimate the models/quantities that I want it to?
- What kind of data (e.g. discrete or continuous) will make my estimation problem easier?

After the experiment

- To complement a traditional analysis (comparing treatments)
- Estimate economically meaningful quantities

What are some things I should be thinking about when estimating structural models?

- How will this help **answer my research question**?
  - Economically meaningful quantities
  - Parameters/quantities that we can interpret
- What are the **underlying assumptions** of the model(s) I take to the data?
- Can I relax these assumptions?
  - More flexible functional forms
  - More parametric models (in mixture or separately)
- How am I accounting for **heterogeneity**?
  - Heterogeneity that I care about
  - Heterogeneity that I don't care about but might affect my estimates

What are some things to think about when doing **Bayesian** work?

- What do my priors say about:
  - Reasonable and unreasonable parameter values?
  - The data I might collect?
  - Economically meaningful quantities
  - The performance of my estimator?
- Data augmentation
  - Good for models with heterogeneity
  - In *Stan*: only continuous variables – integrate the likelihood for discrete heterogeneity

What are some things to think about when implementing a model in *Stan*?

Re-parameterizations

Efficient coding

- Know your bottlenecks!
- Pre-computing things
- Vectorized operations (avoiding `for` loops)

Parallelization: Between- and within-chain

What should I think about when communicating my results?

Pictures really help

Try to communicate both:

- The point estimate (e.g. posterior mean), and
- An expression of uncertainty (e.g. posterior standard deviation, 95% Bayesian credible region)

Beware of over-plotting when it comes to individual-level parameters

Make tables that look comforting to reduced-form frequentists:

- regression-like tables
- Posterior mean with standard deviation in parentheses

Placating the frequentists

Stress the **practical** side of Bayesian estimation: it can turn a difficult maximization problem into a slightly less difficult simulation problem

- Especially for hierarchical models
- Expressions of uncertainty
  - Bayesian – part of the simulation
  - Frequentist – bootstrap or delta method
- **Show them what you can do with Bayes**

Show that you have thought about your priors.

- Have a good answer for why you chose them.
- Maybe do a prior sensitivity analysis

Things not covered today but worth a look

Experiment design

- With the goal of estimating a structural model
- Using structural estimates
- E.g. Bland (2023)

Model selection using cross-validation

- Not specific to Bayes
- Application to QRE here: Bland (2022)

Finite mixture models: more than one data-generating process

Within-chain parallelization in *Stan* (only if you have idle cores)

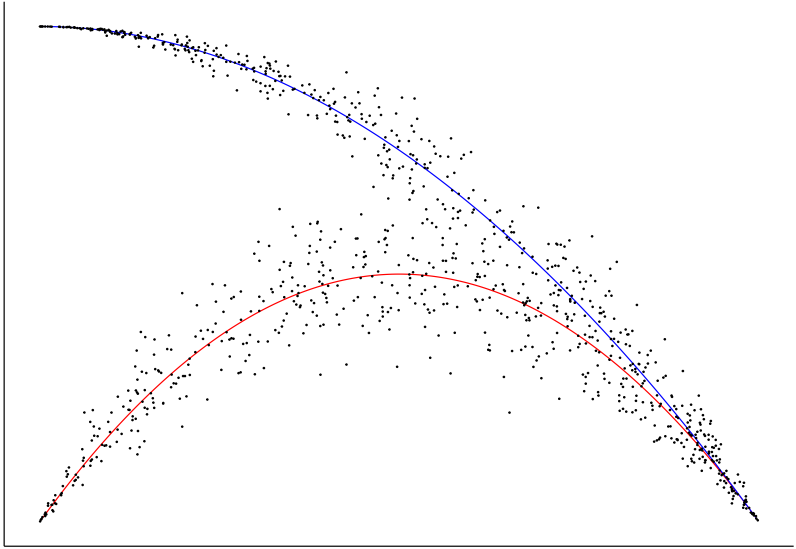
Where are some datasets I can use to practice?

Experimental economics has really good data-sharing norms! If a paper is published, chances are you can get the data.

- Journal websites
- Authors' websites, e.g:
  - Glenn Harrison: https://cear.gsu.edu/gwh/
  - Guillaume Frechette: http://people.cess.fas.nyu.edu/frechette/html/papers.htm
  - Peter Moffatt (Experimetrics online resources): https://www.bloomsburyonlineresources.com/experimetrics-econometrics
- Email people!

`james.bland@utoledo.edu`

Thank you!



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